CSA06507-Generative AI

Lab Experiment's

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Unit 3: Lab Programs for Prompting

Experiment 1: Generate Text Embeddings and Perform Semantic Similarity Search

Aim:

To generate text embeddings for a collection of sentences and perform semantic similarity search to identify sentences that are most similar to a given query using Python.

Algorithm

1. Start the program.
2. Import the required Python libraries.
3. Define a collection of sample sentences.
4. Convert the sentences into numerical text embeddings using a Sentence Transformer model.
5. Accept a search query from the user.
6. Generate an embedding for the search query.
7. Calculate cosine similarity between the query embedding and sentence embeddings.
8. Rank the sentences based on similarity scores.
9. Display the most similar sentences.
10. Stop the program.

Python Program

Google Colab Code:

First, run this installation command in one cell.

Python

Run

!pip install sentence-transformers

Next, run this program in another cell.

Python

Run

Text Embedding and Semantic Similarity Search

from sentence_transformers import SentenceTransformer

from sklearn.metrics.pairwise import cosine_similarity

```
# Step 1: Load the model
model = SentenceTransformer('all-MiniLM-L6-v2')

# Step 2: Sample sentences
sentences = [
    "Artificial Intelligence helps machines learn.",
    "Python is a programming language.",
    "Machine learning is a part of AI.",
    "Students use computers for education.",
    "Deep learning uses neural networks."
]

# Step 3: Generate text embeddings
embeddings = model.encode(sentences)

print("TEXT EMBEDDINGS GENERATED SUCCESSFULLY")

# Step 4: Get search query
query = input("Enter your search query: ")

# Step 5: Generate query embedding
query_embedding = model.encode([query])

# Step 6: Calculate similarity
similarity = cosine_similarity(
    query_embedding, embeddings
)[0]

# Step 7: Sort results
results = sorted(
    zip(sentences, similarity),
```

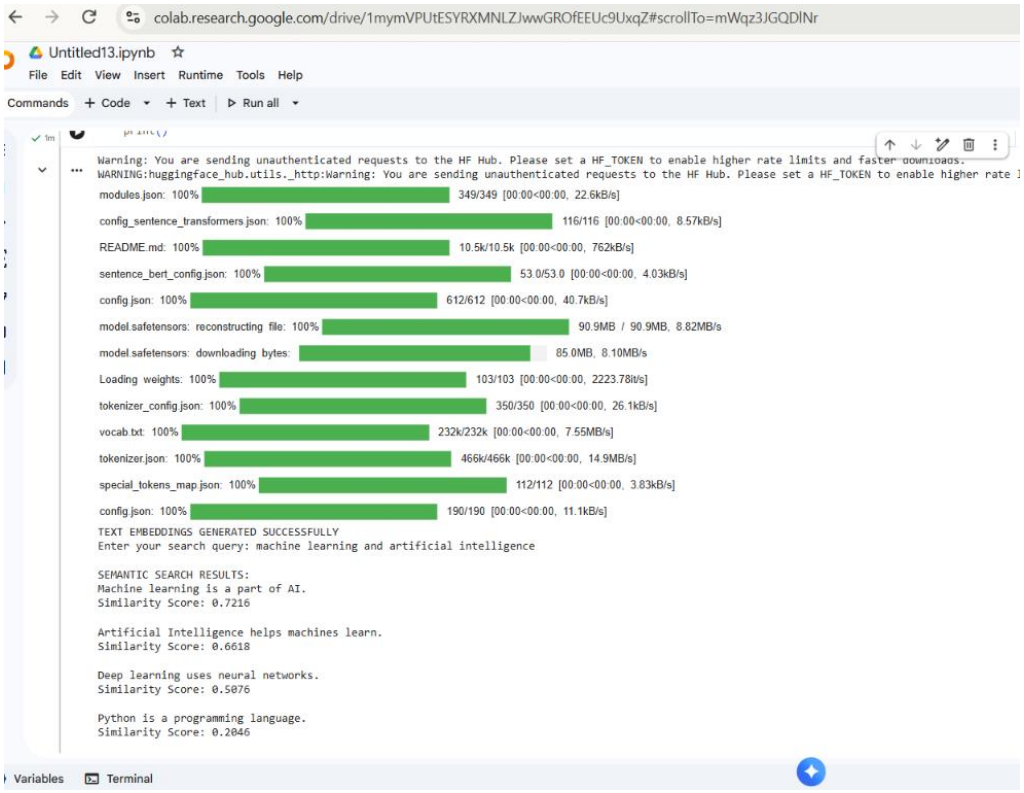
Input

Example input:

Plain text

Enter your search query: Machine learning and artificial intelligence

Output



Explanation

1. Text Embedding:

Text embedding converts sentences into numerical vectors that represent their meaning.

2. Semantic Similarity:

Semantic similarity identifies how closely two sentences are related in meaning.

3. Cosine Similarity:

Cosine similarity measures the similarity between two numerical vectors. A higher score generally indicates greater semantic similarity.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\| A \| \| B \|}$$

Result

Thus, text embeddings were successfully generated using the Sentence Transformer model, and semantic similarity search was performed using cosine similarity. The program identified and displayed the most relevant sentences based on the given search query.